Math for Data Science - Lab 2

Course: Math for Data Science

This lab is about **conditional probability** and **Bayes' rule**, and about what goes wrong when you condition on the wrong thing. We work through two famous courtroom fallacies, simulate the disease-testing example from class, and close with a few exercises from Blitzstein & Hwang.

Setup

Run the cell below once to import everything we need.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Using R instead of Python (optional)

The lab is written in Python, but you can write your answers in R inside this same notebook. This needs R 4.5 or newer, and Jupyter started with rpy2 added: `uv run --with rpy2 jupyter lab`. The `README.md` of the course folder has the details and what to do when it does not work.

If you want to use R instead of Python, uncomment and run the following cell:

```
# %load_ext rpy2.ipython
```

Then, in any subsequent code cell, add `%%R` as the first line to run that cell as R:

```
%%R
x <- rbinom(1000, size = 10, prob = 0.3)
hist(x)
```

Keep running the given Python cells (imports, data loading) as they are. The hints name Python functions, such as `rng.binomial`; you will need to find the R equivalent (`rbinom`) yourself.

Exercise 1: Prosecutor's Fallacy (from Blitzstein & Hwang)

In 1998, Sally Clark was tried for murder after two of her sons died shortly after birth. During the trial, an expert witness for the prosecution testified that the probability of a newborn dying of sudden infant death syndrome (SIDS) was $1/8500$, so the probability of two deaths due to SIDS in one family was $(1/8500)^2$, or about one in 73 million. Therefore, he continued, the probability of Clark's innocence was one in 73 million.

1. Is this reasoning correct? If not, describe the major problems that you see.

here your solution

2. Which conditional probability is the expert using in his reasoning? Which other conditional probability would you consider? Explain why the difference is important.

here your solution

3. Let's put numbers on the intuition above. Suppose double infanticide is *also* rare: say the prior probability that a mother in this situation is guilty is $P(G) = 1/100,000$. Take the expert's figure at face value, $P(\text{evidence} \mid \text{innocence}) = (1/8500)^2$, and suppose that a guilty mother would certainly produce this evidence, $P(\text{evidence} \mid \text{guilt}) = 1$. Compute $P(\text{innocence} \mid \text{evidence})$.

> **Note:** the point of this exercise is not the exact number - the inputs are all debatable - but that a quantity the expert claimed was "one in 73 million" is in fact nowhere near that.

```
# here your code
```

Exercise 2: Disease Testing Simulation

We will now simulate the disease-testing example from class.

1. First, create an array called `flu` that determines whether every member of a population of size 100,000 is healthy or sick. Let the probability of being sick be 1%.

```
# here your code
```

2. Now create an array called `test` that determines whether the person tested positive. Let the **sensitivity** of the test (the probability of identifying sick individuals as positive) be 0.95, and the **specificity** (the probability of identifying healthy individuals as negative) be 0.95 as well. Put `flu` and `test` into the same data frame. Add a column called `accurate` that is `True` when the test is right and `False` when it is wrong.

```
# here your code
```

3. What is the accuracy of the test? What is the probability of being sick given a positive result? What is the probability of being healthy given a negative result? How does this compare to what you derived mathematically using Bayes' rule?

> **Tip:** `value_counts(normalize=True)` is the pandas equivalent of R's `prop.table(table(...))`.

```
# here your code
```

4. Suppose the company that designed the test has the opportunity to raise the specificity, while keeping the sensitivity fixed at 0.95. Plot the relationship between the specificity of the test and the probability of being sick given a positive result over the region from 0.95 to 1 (use intervals of 0.001). How should the company invest in raising specificity to yield the highest returns?

```
# here your code
```

here your solution

5. Now suppose there is a cost to raising specificity: for any amount by which you raise specificity, you lose the same amount of sensitivity. Put this situation into the same plot as in (4). Should the company still pursue boosting specificity under these conditions? What if you lost *triple* the sensitivity?

```
# here your code
```

here your solution

Exercise 3: Defence Attorney's Fallacy (from Blitzstein & Hwang)

A woman has been murdered, and her husband is put on trial for this crime. Evidence comes to light that the defendant had a history of abusing his wife. The defense attorney argues that the evidence of abuse should be excluded on grounds of irrelevance, since only 1 in 10,000 men who abuse their wives subsequently murder them. Should the judge grant the defense attorney's motion to bar this evidence from trial?

Suppose the defense attorney's 1-in-10,000 figure is correct, and further assume: 1 in 10 men commit abuse against their wives; 1 in 5 married women who are murdered are murdered by their husbands; and 50% of husbands who murder their wives previously abused them. Also assume that if the husband of a murdered wife is *not* guilty of the murder, then the probability that he abused her reverts to the unconditional probability of abuse.

1. Name the main events in this problem. Using those events, describe the probability behind the defense's argument.

here your solution

2. Do you think that the probability considered by the defense attorney is the relevant one? Why?

here your solution

3. Which probability would you calculate for this trial? Calculate it - first by hand, then verify it in code.

here your solution

```
# here your code
```

Exercise 4: Crime and Blood Type

This is Exercise 25 (conditional probability) from Blitzstein & Hwang.

A crime is committed by one of two suspects, A and B. Initially, there is equal evidence against both of them. In further investigation at the crime scene, it is found that the guilty party had a blood type found in 10% of the population. Suspect A *does* match this blood type, whereas the blood type of Suspect B is unknown.

1. Given this new information, what is the probability that A is the guilty party?

2. Given this new information, what is the probability that B's blood type matches that found at the crime scene?

here your solution

```
# here your code
```

Exercise 5: Children's Age

Exercise 30 (conditional probability) from Blitzstein & Hwang.

A family has 3 children, creatively named A, B and C.

1. Discuss intuitively (but clearly) whether the event "A is older than B" is independent of the event "A is older than C".

2. Find the probability that A is older than B, given that A is older than C.

here your solution

3. Verify part 2 by simulation: draw many random birth orders and estimate the conditional probability directly.

```
# here your code
```

Exercise 6: The Chess Game

Exercise 35 (conditional probability) from Blitzstein & Hwang.

You are going to play 2 games of chess with an opponent whom you have never played before. Your opponent is equally likely to be a beginner, intermediate, or a master. Depending on which, your chances of winning an individual game are 90%, 50%, or 30%, respectively.

1. What is your probability of winning the first game?

2. Congratulations: you won the first game! Given this information, what is the probability that you will also win the second game (assume that, given the skill level of your opponent, the outcomes of the games are independent)?

3. Explain the distinction between assuming that the outcomes of the games are independent and assuming that they are *conditionally* independent given the opponent's skill level. Which of these assumptions seems more reasonable, and why?

here your solution

Exercise 7: The Driver Type

Exercise 34 (conditional probability) from Blitzstein & Hwang.

Suppose there are two types of drivers: good drivers and bad drivers. Let G be the event that a certain man is a good driver, A the event that he gets into a car accident next year, and B the event that he gets into a car accident the following year. Let $P(G) = g$ and $P(A \mid G) = P(B \mid G) = p_1$, $P(A \mid G^c) = P(B \mid G^c) = p_2$, with $p_1 < p_2$.

Suppose that given the information of whether or not the man is a good driver, A and B are independent.

1. Explain intuitively whether or not A and B are independent.

2. Find $P(G \mid A^c)$.

3. Find $P(B \mid A^c)$.

here your solution

4. Write the two formulas above as Python functions and check them against a simulation with $g = 0.7$, $p_1 = 0.05$, $p_2 = 0.3$. Also compare $P(B \mid A^c)$ to the unconditional $P(B)$.

```
# here your code
```